

indicating that the pairs data contain significantly more noise than the discrete-choice data. This result is not surprising because of the more cognitively burdensome nature of the pairs task.

WILLINGNESS-TO-PAY ESTIMATES

Because the joint model captures all the available information on subjects' preferences in one set of parameters, we use this model to estimate WTP. As discussed above, we can convert the parameters of the utility index shown in the joint model into a dollar metric by rescaling the index using the marginal utility of money. Employing SP techniques allows us to recover WTP estimates for all relevant combinations of symptoms and activity levels in the experimental design.ⁿⁿ For example, using Equation (13), the average WTP to avoid 1 day of coughing, wheezing and shortness of breath, which confines subjects to their homes but allows them to care for themselves, is calculated as follows:

$$\begin{aligned} \text{WTP}(\text{BREATH, ATHOME, Days}) &= \frac{\text{Current Utility (Days)} - \text{Utility (BREATH, ATHOME, Days)}}{\text{Marginal utility of money}} \\ &= \frac{\ln(1 + 1) \cdot V_{\text{Current}} - [\ln(1 + 1) \cdot \beta_{\text{BREATH}} + \beta_{\text{ATHOME}}]}{\frac{1}{2} \cdot \left[\frac{-dV}{d\text{Cost}^{\text{CH}}} + \frac{-dV}{d\text{Cost}^{\text{GP}}} \right]} \\ &= \frac{0.6931 \cdot 0.2955 - [0.6931 \cdot (-0.0294) + (0.0225)]}{\frac{1}{2} \cdot \left[\frac{0.01975}{2 \cdot \sqrt{375}} + \frac{0.05859}{2 \cdot \sqrt{202}} \right]} = 158, \end{aligned}$$

where CH and GP refer to choice and graded-pair estimates, respectively, and cost is averaged over all the questions in the each question format where these two symptom and activity levels appear together.^{oo} The bootstrapped 90% confidence interval is CAN\$100–CAN\$225. The confidence interval is obtained by drawing 1000 times from the multivariate normal distribution of coefficients and their variance–covariance matrix. The foregoing calculation is performed for each draw, and the confidence interval lower and upper bounds of the WTP estimates are the 5th and 95th percentile values, respectively.

Table 6 shows WTP estimates and 90% confidence intervals to avoid 1-day, 5-day and 10-day episodes of the relevant symptom and activity level combinations, given subjects' baseline health for the past year. The first column of estimates shows WTP estimates for the MILD activity restriction, which combines SOMELIM and NOSOC. We show zero WTP for 1-day episodes of NOSE, EYE and BREATH. The point estimates for these symptoms are negative, meaning that the estimated average baseline current health in our sample actually was less than the estimated utility for these outcomes. Moving from these health states to current health would reduce welfare, hence the negative point estimate for WTP. This result is not surprising considering that approximately 36% of our 344 subjects have experienced cardiovascular or respiratory conditions within the past year. In addition, non-symptomatic subjects may suffer from a variety of other ailments. Thus, the results show that on average this sample of subjects is not willing to pay to avoid an episode of these mild conditions.

Examining the WTP estimates across symptoms shows that there are essentially two levels of WTP estimates—relatively low numbers associated with the milder symptoms of NOSE and EYE and higher numbers associated with the other five symptoms. These results reflect the variation in the symptom coefficients in the joint estimates. Table 6 also shows that as activity restrictions increase for a given symptom (i.e. as one moves to the right from 'no limitations' to 'in hospital'), WTP also increases. Differences between WTP estimates in adjacent activity levels generally are not statistically significant.

Table 6. WTP estimates for each symptom/activity level combination for 1-day, 5-day and 10-day episodes (1997 CAN\$)

Symptoms	Duration (days)	Daily activity levels			
		MILD ^a	ATHOME	NEEDHELP	INHOSP
NOSE	1	0 ^b (-168/-83)	3 (-50/57)	— ^c	— ^c
	5	0 ^b (-204/4)	29 (-89/148)	— ^c	— ^c
	10	0 ^b (-225/53)	44 (-110/199)	— ^c	— ^c
EYE	1	0 ^b (-119/-32)	54 (-2/108)	— ^c	— ^c
	5	29 (-76/138)	162 (44/282)	— ^c	— ^c
	10	85 (-53/232)	222 (67/379)	— ^c	— ^c
FLUTTER	1	24 (-26/73)	171 (114/236)	288 (232/359)	427 (352/517)
	5	310 (192/438)	468 (341/615)	588 (462/741)	704 (568/870)
	10	468 (309/641)	631 (466/825)	754 (589/955)	857 (684/1065)
BREATH	1	0 ^b (-60/49)	158 (100/225)	286 (224/363)	448 (371/536)
	5	266 (141/405)	435 (299/589)	566 (427/732)	712 (566/872)
	10	415 (249/602)	589 (411/789)	721 (537/937)	857 (668/1058)
ACHE	1	26 (-36/86)	199 (133/275)	335 (267/420)	513 (426/623)
	5	362 (214/523)	544 (382/722)	682 (529/866)	845 (677/1046)
	10	547 (350/762)	734 (524/967)	873 (668/1114)	1027 (813/1278)
SWELL	1	56 (-1/117)	229 (164/306)	365 (295/452)	535 (443/644)
	5	439 (296/598)	621 (469/799)	761 (598/954)	908 (737/1111)
	10	650 (459/869)	837 (634/1073)	979 (767/1228)	1114 (896/1375)
PAIN	1	14 (-48/75)	190 (125/263)	329 (260/411)	510 (426/611)
	5	338 (193/489)	522 (370/693)	663 (508/841)	827 (665/1015)
	10	516 (322/719)	705 (503/934)	848 (645/1084)	1002 (791/1240)

^a MILD combines SOMELIM and NOSOC.^b Negative point estimate shown as zero WTP. See text for interpretation.^c These combinations fall outside the scope of the experimental design.

Overall, WTP estimates show the most dramatic differences across activity levels, and while there are differences among the symptoms, these differences appear to be less salient to subjects for a given activity limitation.

CONCLUSIONS

Using two different SP elicitation formats is a significant improvement over previous health valuation studies. The two formats collect slightly different information and induce subjects to employ different evaluation strategies. Thus, combining both formats allows us to capitalize on the information provided by each format and on the different cognitive processes that subjects use in each situation. The two formats also differ in their formal utility theoretic basis. The graded-pair format elicits values for marginal trade-offs among health states, whereas the discrete-choice format elicits total values to avoid a given condition relative to the subject's current health. Thus, each elicitation format offers certain advantages, and we believe that employing both formats provides more robust and valid estimates.

Comparisons of WTP estimates from the joint model with the few published estimates for similar conditions based on conventional CV methodology indicates our estimates are not systematically larger or smaller [15]. However, unlike other studies in the WTP literature, our SP estimates are derived from complete multi-attribute health state descriptions that are clearly specified and consistent across the same sample of subjects. Thus, we are able to explicitly account for the separate effects of symptoms, duration, activity restrictions, and current health on WTP. The many possible combinations of these attributes correspond to a broad range of cardiovascular and respiratory conditions. This approach allows us to recover consistently derived WTP estimates for these conditions and facilitates matching outcome values more accurately than previously was possible, to evaluate various health care interventions and policies.

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NOTES

- a. Terminology has not been standardized among various disciplines. In this study, we use the term 'stated-preference' to refer to a group of techniques used primarily in market research studies to measure consumer preferences. The term 'conjoint analysis' also has been used to describe some of these techniques. Although CV also could be called an SP technique, CV was developed independently by environmental economists and generally relies on a different set of elicitation formats and analytical approaches.
- b. In the resource economics literature, Gan and Luzar [23] use SP to value hunting trips in Louisiana. Mackenzie [18] values hunting trips in Delaware using SP analysis. Opaluch *et al.* [24] also use SP to describe public preferences for siting a noxious facility. Adamowicz *et al.* [25,26] use SP to explain recreational site choice selection. Johnson *et al.* [27] use SP to estimate electric customers' WTP for environmental and other attributes of electricity generation. Roe *et al.* [28] use SP to value the effects on sport fishing of implementing alternative management plans to restore runs of Atlantic salmon in Maine.
- c. Preference elicitation for assessing welfare changes requires that subject responses conform to basic postulates of utility theory. Among other things, this rules out lexicographic preferences, where subjects refuse attribute trade-offs. SP data often facilitate tests of conformity with utility theoretic requirements [29]. Failure to accept trade-offs often arises because the range of attribute levels offered is not salient. Ensuring appropriate attribute ranges is a basic task of survey development and testing.
- d. Defining the properties of such preferences has been explored by multi-attribute utility theory [30]. For an application to health state utility, see Torrance *et al.* [31].
- e. A third format, *ranking*, has also been used in market research. For a more extensive review of experimental design possibilities, see Green *et al.* [32].

- f. For a review of choice experiments in marketing applications, see Carson *et al.* [33].
- g. This approach has been used in modelling travel behaviour, evaluating new products and estimating recreation demands. See, for example, Louviere and Hensher [34]. Health studies that use this approach include Bryan *et al.* [9], Propper [11] and Ryan and Hughes [8].
- h. See also Huber *et al.* [13], Huber [35], Tversky *et al.* [36], Payne [37,38], Huber and Klein [39], Elrod *et al.* [40], Akaah [41] and Huber and Holbrook [42] for discussions of the different cognitive processes involved in graded-pair versus choice tasks.
- i. We initially treated the QWB function states as separate attributes, but we found in pre-testing that subjects tended to view these function states as interdependent, which complicates the development of an orthogonal experimental design.
- j. The range of cost values offered do not determine higher or lower WTP values. Rather, variations in cost induce trade-offs among attributes. These trade-offs reveal information about marginal utilities, including the marginal utility of money. Calculating WTP using estimated parameters is discussed below.
- k. To avoid introducing correlation into the design by simply deleting combinations that violated constraints, we constructed a series of orthogonal designs for groups of constrained attribute levels, then concatenated the sub-designs into a candidate design from which a design of the desired size and efficiency was drawn.
- l. Thus, NOSE and EYE are never seen with ATH-OME, NEEDHELP and INHOSP, as well as values of \$200 and \$500.
- m. The payment vehicle for both formats is described as illness-related costs that are not covered by the government health system or a company insurance plan. These costs are associated with items that reduce discomfort or the length of illness (such as vitamins, medicines, air filters or humidifiers, special foods or liquids, or other optional treatments). These costs may also include such costs as child care, while sick, or transportation to the doctor. Subjects were instructed to assume that any missed time from work would be covered by paid sick leave.
- n. The choice format was added to the study after a large-scale pre-test of the graded-pairs format. Peer reviewers suggested that adding a graded-pairs component to the study would provide better ability to control for current health status, as well as offer the benefits of multiple-format elicitations discussed above. For details on the evolution of the experimental design, see Johnson *et al.* [15,44] and Desvousges *et al.* [43].
- o. These illnesses included minor lung illnesses, lung infections, chronic lung conditions, coronary artery disease, heart failure and heart rhythm problems.
- p. The computer format was an advantage in this part of the survey because if subjects answered the quiz question incorrectly, the computer provided them with the correct response.
- q. Although we did not actively recruit subjects who had ever experienced the heart and lung illnesses of interest, 43.6% of the sample used in the analysis have been diagnosed with one of these conditions at some point in their lives.
- r. The first social science application of this approach was by McKelvey and Zavoina [45] (see Elrod *et al.* [40], Akaah [41] and Huber and Holbrook [42]). Roe *et al.* [28] and Johnson and Desvousges [46] discuss parameter estimation and WTP calculations for graded-pairs data.
- s. See Keeney and Raiffa [30] for an analysis of the properties of such utility functions.
- t. The square root form was chosen over linear or log by estimating a Box-Cox model.
- u. This procedure assumes that subjects have no systematic preference for screen location.
- v. Because the original response scale indicates which profile is preferred and how much it is preferred, this rearrangement maps response 3 into 5, 2 into 6 and 1 into 7. Response 7 indicates maximum utility difference and 4 indicates indifference; so C'_i equals the recoded response minus 4. We also have estimated binary models (not reported here), which suppress the intensity of preference information. Such models fit the data poorly relative to the ordered probit specification. Also, it is not clear how to classify indifference in a binary model.
- w. The maximum likelihood procedure used to estimate the model parameters normalizes the α_0 threshold at $-\infty$ and α_5 at $+\infty$, and does not include an intercept term.
- x. The way to generalize μ in a probit-type model is to retain the normalization for the base category that $\mu_0^i = 1$. For all other categories, $\mu_k^i = \mu^i$ is modelled as a function of individual characteristics. This guarantees that the estimated value of μ will be positive for every observation and that α_k adjusts to account for the normalization of the base category. Since ρ is the normalized intra-subject correlation, we have controlled for both intra-subject and inter-subject heteroskedasticity, which removes bias in estimated standard errors from both sources. Our ability to separately identify these two sources of heteroskedasticity requires that we assume independence in of the permanent and transitory components of the error term. If these errors are correlated, then the intra-subject and inter-subject variances are confounded to an unknown extent.
- y. This specification is a generalization of Butler and Moffitt's [47] derivation for binary probit.
- z. The basic exposition of the properties of this model can be found in McFadden [48].